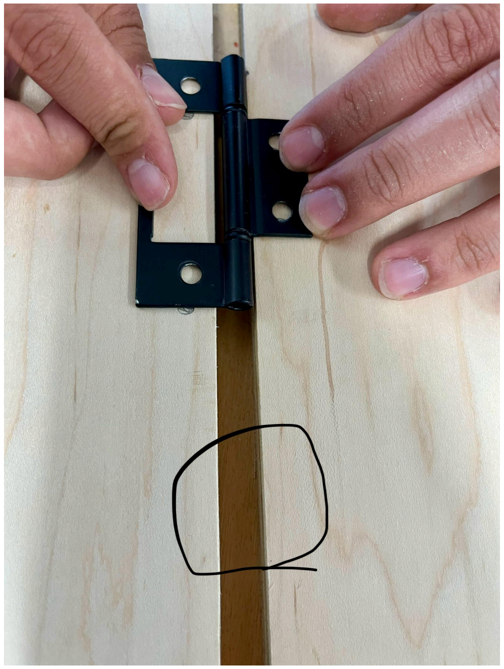
My Design: Adaptive Technology
Armin Zarbaft, Jared Wagner, John Estock

Element H: Prototype Testing and Data Collection

Testing Plan: After constructing our prototype, we wanted to ensure that it would work when being installed in the middle island. For this, we devised a plan to test our prototype.
To make our door, we needed to cut out parts of our door from wood and make all the pieces of wood uniform in size. From this, we had two extra pieces of wood that had not been stained, but were of the same dimensions and material of the wood we plan on using for the actual installation.
With this extra wood, we will drill the bifold hinges in to make sure that the hinges could support the weight of the door, and that there are no problems with the bifold hinges allowing the two pieces of wood to fold onto itself successfully.

Another item of concern we wanted to test for was the length of the screws for the bifold hinges. The length of the screws are 3/4 inch which matches the width of our wood. We had to avoid the possibility of the screw splintering into the bottom of our wood, so we will test it by screwing the screws with the hinges into our extra pieces of wood.

To collect data and results of our testing, we will take note of and write down any significant results, findings, or changes to metrics. One measurement we need to test for is the distance between the two pieces of wood when the hinges are screwed in while flat. This space holds the middle of the hinge, and we plan on testing for which distance between the wood allows for the best smooth fold. We wanted to test ¼ inch, ⅜ inches, and if necessary, 0 inches or no gap at all. We really think we need a gap, so we are saving it for last in hopes that one of the first two distances will allow for proper fold. The gap can be seen circled in the photo below.



Expert Verification of Testing Plan:

Nathan Johnson is a vehicle engineering associate at *Exponent*, a global engineering and scientific consulting firm. He has experience with post car collision analysis, and other various components of engineering, math, and science.

We have contacted him in regards to our testing plan and are waiting for feedback.